

MODULE 8

ANALOG SIGNAL READING, SCALING AND BASIC PID CONTROL

Siemens TIA Portal PLC Course

Module 8: ANALOG SIGNAL READING, SCALING AND BASIC PID CONTROL

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Software: Siemens TIA Portal

Language: Ladder Diagram (LAD)

Introduction

Most modern industrial processes require the measurement and control of continuous variables such as temperature, pressure, flow, level, speed, and position. These variables are acquired through analog signals coming from field sensors and instruments.

Siemens PLCs allow these signals to be processed through analog input and output modules, perform the corresponding scaling, and apply PID control algorithms to automatically regulate industrial processes.

In this module you will learn to read 4-20 mA and 0-10 V analog signals, convert them into engineering units, and perform basic PID control using TIA Portal V20.

Analog Signal

An analog signal is an electrical signal whose value can vary continuously within a given range.

Unlike digital signals, analog signals can take not only the states 0 or 1, but any intermediate value.

Common Industrial Variables

- Temperature
- Pressure
- Level
- Flow
- Speed
- Position
- Weight
- Humidity

Types of Industrial Analog Signals

The most commonly used signals in industry are:

- 0-10 V
- 1-5 V
- 0-20 mA
- 4-20 mA

0-10 V Analog Signal

Definition

The 0-10 V signal uses a voltage proportional to the measured variable.

Example:

Voltage, Temperature

0 V, 0 °C

10 V, 100 °C

Industrial Applications

- Variable frequency drives
- Position sensors
- Potentiometers
- Speed control
- Electronic instrumentation

4-20 mA Analog Signal

Definition

The 4-20 mA signal uses an electrical current proportional to the measured variable.

Example:

Current, Level

4 mA, 0 %

20 mA, 100 %

Advantages of the 4-20 mA Standard

- Greater immunity to electrical noise
- Allows long wiring distances
- Higher accuracy
- Easy fault detection

Fault Detection

If the PLC receives 0 mA, or a value lower than 4 mA, it is possible to detect:

- Cut cable
- Disconnected sensor
- Transmitter failure

For this reason, the 4-20 mA standard is the most widely used in industry.

Reading Analog Signals in Siemens PLCs

Siemens analog inputs convert the electrical signal into a digital value.

For example, a typical conversion range:

0 - 27648
or
5530 - 27648
for 4 - 20 mA signals

depending on the module used.

Reading Example

Temperature sensor: 0 - 100 °C
Value read by the PLC: 13824

The value does not yet represent degrees Celsius, but rather a digital representation of the analog signal. Therefore, scaling is required.

Scaling of Analog Signals

Definition

Scaling consists of converting the digital value delivered by the analog module into an understandable engineering unit.

Examples: °C, bar, liters, meters, percentage, RPM

General Scaling Formula

$$\begin{aligned} \text{Scaled Value} = & \\ & ((\text{Input Value} - \text{Minimum Input}) \\ & \times (\text{Maximum Output} - \text{Minimum Output}) \\ & \div (\text{Maximum Input} - \text{Minimum Input})) \\ & + \text{Minimum Output} \end{aligned}$$

Practical Example

Temperature transmitter: 0 - 100 °C
Analog input: 0 - 27648
Value read: 13824
Result: 50 °C

Scaling Applications

- Temperature
- Level
- Pressure
- Flow
- Speed
- Position

Scaling Blocks in TIA Portal

Siemens includes blocks that facilitate the scaling process.

Among the most commonly used:

- NORM_X
- SCALE_X

NORM_X Block

Converts an input value into a normalized range: 0.0 – 1.0

SCALE_X Block

Converts the normalized value into an engineering unit.

Example: 0 – 100 °C

Scaling Example

Analog Input
↓
NORM_X
↓
SCALE_X
↓
Temperature = 75 °C

What Is a PID Controller?

PID stands for:

P = Proportional
I = Integral
D = Derivative

It is a mathematical algorithm used to keep a process variable at a desired value.

PID Control Variables

SP (Set Point): Desired value of the process.

PV (Process Variable): Actual measured value.

CV (Control Variable): Controller output.

Error

$$\text{Error} = \text{SP} - \text{PV}$$

Proportional Action (P)

The output is proportional to the existing error.

Greater error: greater control action.

Integral Action (I)

Eliminates permanent error by accumulating the error over time.

Derivative Action (D)

Anticipates future changes by observing the rate of change of the error.

Industrial Applications of PID

- Temperature control
- Pressure control
- Level control
- Speed control
- Flow control
- Industrial furnaces
- Variable frequency drives
- Pumping systems

Practical PID Control Example

Furnace temperature control.

Set Point: 80 °C

Current temperature: 65 °C

Error: 15 °C

The PID controller increases the heating power until the desired value is reached.

PID Blocks in Siemens

TIA Portal includes technology blocks to implement PID control:

- PID Compact
- PID Temperature
- Technology Objects

Basic PID Parameters

Proportional Gain (Kp)

Determines the intensity of the response.

Integral Time (Ti)

Eliminates steady-state error.

Derivative Time (Td)

Reduces overshoot and improves stability.

Programming Best Practices

- Check the analog module configuration
- Use the correct signal type
- Always scale signals before using them
- Filter signals affected by electrical noise
- Check the operating range of the sensors
- Carefully adjust the PID parameters
- Run simulation tests before commissioning

Practical Exercise

Design a program that allows you to:

Exercise 1

Read a level transmitter: 4 – 20 mA
and convert it to: 0 – 100 %

Exercise 2

Read a temperature sensor: 0 – 10 V
and scale it to: 0 – 150 °C

Exercise 3

Implement a basic PID control to keep a furnace temperature at: 80 °C
using a heater controlled by the PLC.

Conclusion

Analog signals and PID control are fundamental elements of modern industrial automation. The correct reading of 4-20 mA and 0-10 V signals, together with proper scaling and the implementation of PID control strategies, enables the development of precise, reliable, and efficient control systems.

Mastering these tools is an essential step for any Siemens PLC programmer who wants to develop advanced industrial applications using TIA Portal V20.